

REED PUTNAM PRACTICE · FALL 2026

Inequalities & Analysis

Week 6 · guess where equality holds, then prove it

Thu Nov 5 / Fri Nov 6

The toolbox (1)

The habit

Before proving any bound, **guess where equality holds**: all variables equal, a boundary point, a bang-bang $f = \pm 1$. The guess picks the inequality and its weights — and “how large can it be” is incomplete without the example.

- > **AM–GM**: $\frac{x_1 + \dots + x_n}{n} \geq (x_1 \cdots x_n)^{1/n}$ for $x_i \geq 0$; rescale so equality is “all equal.”
- > **Cauchy–Schwarz**: $(\sum a_i b_i)^2 \leq \sum a_i^2 \sum b_i^2$; Engel form $\sum \frac{a_i^2}{b_i} \geq \frac{(\sum a_i)^2}{\sum b_i}$.
- > **Jensen**: $\varphi'' \geq 0 \Rightarrow \sum w_i \varphi(x_i) \geq \varphi(\sum w_i x_i)$; if not convex, the **tangent line trick**.

The toolbox (2)

- > **Integrals:** bound integrands pointwise, splitting where a sign changes. Absorb $\int f = c$ via $\int fw = \int f(w - \lambda) + \lambda c$, with λ chosen so $w - \lambda$ changes sign at the switch.
- > **Sequences:** monotone and bounded $\Rightarrow$ convergent; then the limit solves the fixed-point equation. **Telescope:** $x_n = F(x_n) - F(x_{n+1})$. Substitute $(b_n = e^{a_n}, 1/a_n)$ to get " $b_{n+1} \approx b_n + 1$."
- > **Rolle / MVT:** k zeros of f give $k - 1$ of f' . For $f + \lambda^{-1}f'$, use $(e^{\lambda x}f)' = \lambda e^{\lambda x}(f + \lambda^{-1}f')$.
- > **Grönwall:** $|k'| \leq Ck, k \geq 0 \Rightarrow (e^{Cx}k)' \geq 0$, so $k(x) \geq k(0)e^{-Cx}$.

Example 1

Telescoping and limits (Putnam 2016 B1)

Let $x_0 = 1$ and $x_{n+1} = \ln(e^{x_n} - x_n)$. Show that $\sum_{n \geq 0} x_n$ converges and find its sum.

→ work it on the board (~4 min)

Example 2

Rolle with an exponential multiplier (Putnam 2015 B1)

If f is three times differentiable on $\mathbb{R}$ with at least five distinct real zeros, then

$$f + 6f' + 12f'' + 8f'''$$

has at least two distinct real zeros.

→ work it on the board (~3 min)

Example 3

Tangent line trick

For $a, b, c > 0$ with $a + b + c = 3$, show that

$$\frac{1}{a^2 + 1} + \frac{1}{b^2 + 1} + \frac{1}{c^2 + 1} \geq \frac{3}{2}.$$

→ work it on the board (~2 min)

How to recognize it

- > Symmetric expression, constraint $\sum x_i = 1$: AM–GM, Cauchy–Schwarz, Jensen; equality case first.
- > “How large can $\int fw$ be,” f bounded, one constraint: bang-bang $f = \pm 1$, one switch.
- > A recurrence and $a_n - \log n$ or $\sqrt{n} a_n$: asymptotics; compare with $y' = F(y)$.
- > $f(0) = f(1) = 0$, or $|f'|$ bounded by $|f|$: Rolle, MVT, Grönwall; build $e^{\lambda x} f$ or $f^2 + g^2$.

Pitfalls

- > AM–GM needs nonnegative terms; Jensen needs convexity on the whole range.
- > “Let $L = \lim a_n$ ” is a candidate, not a proof: show convergence first.

Now you

Problems 1–4.

1–2 warm-ups · 3–4 easy Putnam · 5–6 take-home

35 minutes, groups of 2–3

Solutions posted after Friday.